

Problem 1

- Need to minimize the following:

$$C(\boldsymbol{\omega}) = -\frac{1}{N}L(\boldsymbol{\omega}) + \lambda \boldsymbol{\omega}^T \boldsymbol{\omega}$$

where

$$L(\boldsymbol{\omega}) = \sum_{j=1}^n y_j(w_0 + \sum_{i=1}^p w_i x_{ji}) - \ln(1 + \exp(w_0 + \sum_{i=1}^p w_i x_{ji}))$$

We minimize this loss using gradient descent, using:

$$\boldsymbol{\omega}^{t+1} \leftarrow \boldsymbol{\omega}^t - \eta \lambda \boldsymbol{\omega}^t + \frac{\eta}{N} \frac{\partial L(\boldsymbol{\omega}^t)}{\partial \boldsymbol{\omega}}$$

For each of the three datasets, we do the same thing. We first find a learning rate η that permit the loss to converge in at most 200 iterations. We loop over a grid to find the best η (in terms of smallest final training lose), and obtain the plots shown in figure 1. We then report the misclassification error rates for the train and test datasets for the three datasets. This is reported in 1.

dataset	best eta	train misclass	test misclass
Gisette	0.200000	0.000000	0.020000
Madelon	1.000000	0.259500	0.426667
Dexter	0.200000	0.000000	0.140000

Table 1: Best learning rate η , training and test misclassification error, Logistic Regression.

Lastly, we are asked to plot the ROC curve of both the training and test sets. We do so in figure 2

Problem 2

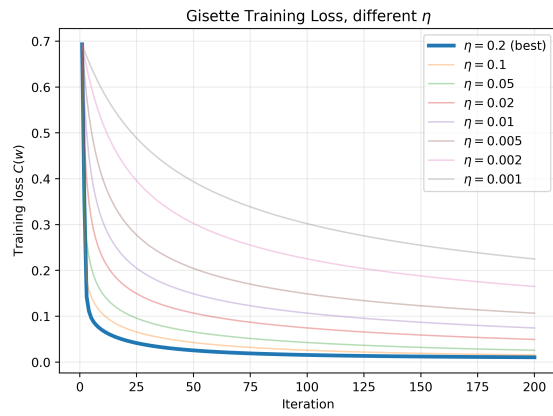
We need to minimize analytically:

$$C(\boldsymbol{\omega}) = \frac{1}{N} \sum_{i=1}^n (y_i - \boldsymbol{\omega}^T \mathbf{x}_i - \omega_0)^2 + \lambda (\boldsymbol{\omega}^T \boldsymbol{\omega} + \omega_0^2)$$

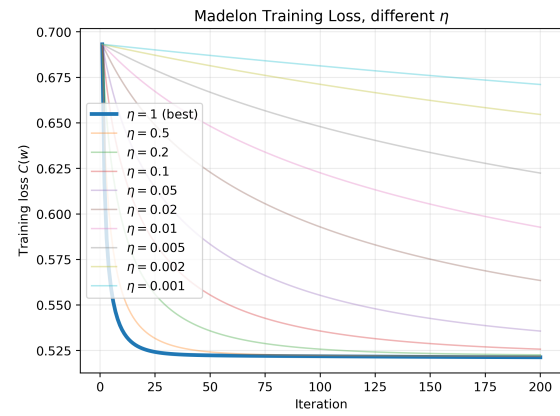
where $\lambda = 0.001$, $(\mathbf{x}_i, y_i)$ are the training data, and $y_i \in \{-1, +1\}$. Predict classification via $\hat{y} = \text{sign}(\boldsymbol{\omega}^T \mathbf{x} + \omega_0)$.

- We can report the misclassification error of the training and test datasets in the following table:

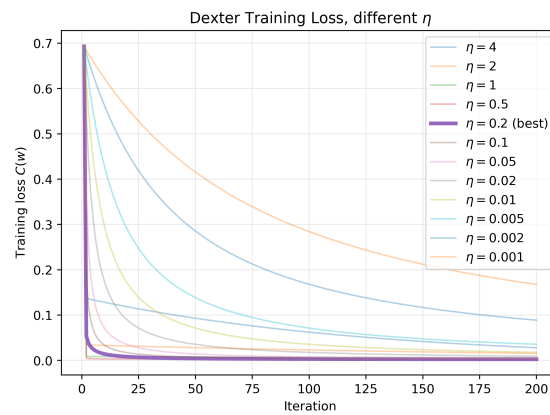
The train misclassification error rate is the same (0.00), though the test error rate is higher (0.091 vs. 0.020).



(a) Top left



(b) Top right

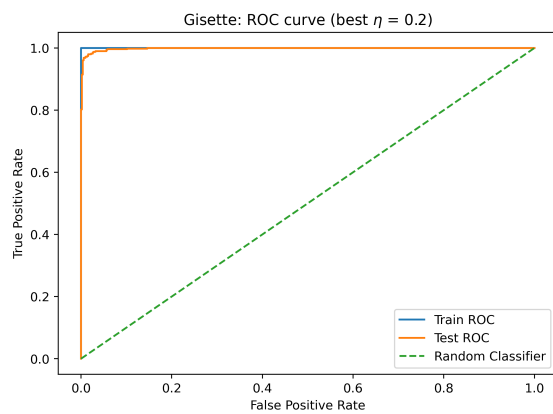


(c) Bottom

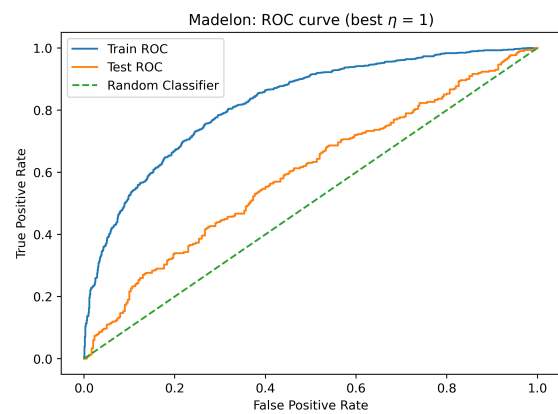
Figure 1: Two images on top, one below.

train misclass error	test misclass error
0.000000	0.091000

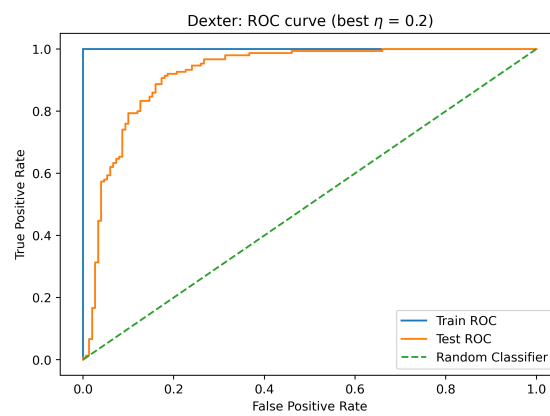
Table 2: misclassification Error for training and test datasets, Ridge like regression



(a) Top left



(b) Top right



(c) Bottom

Figure 2: Two images on top, one below.

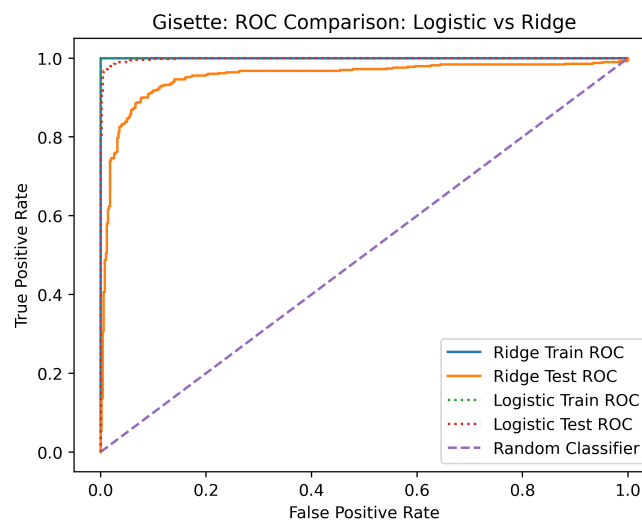


Figure 3: abcd.

- We can display the ROC curve for both the ridge and logistic models in figure 3.

The train ROC for the logistic and ridge models are practically the same. The ROC for the logistic model is better than the ridge though; tighter to the upper left hand corner.